

AP CSP | Lesson 09: Expressions & Strings

Homework | Multiple-choice and written practice

Name: _____ Class: _____ Date: _____

Time: 40 minutes | Total: 30 teacher marks (not an AP score) | Show working.

Original AP-style MCQs and classroom written practice; these are not official AP questions.

Q5: select exactly TWO answers. CONCAT joins two strings with no added spaces. SUBSTRING(s,start,length) uses 1-based positions and a character count. These are supplied teaching procedures.

1. After $x \leftarrow 6$; $y \leftarrow x$; $x \leftarrow 10$, what value does y hold? [1 mark]

A. 6
B. 10
C. 16
D. The expression x , which updates automatically

Answer: _____

2. What is the value of $4 + 3 * 5$? [1 mark]

A. 35
B. 19
C. 27
D. 60

Answer: _____

3. In AP pseudocode, what is $13 / 4$? [1 mark]

A. 3
B. 1
C. 3.25
D. 4

Answer: _____

4. A program evaluates $23 \text{ MOD } 6$. What value is produced? [1 mark]

A. 3
B. 3.8333...
C. 6
D. 5

Answer: _____

5. Using the supplied procedures above, which TWO statements are true? Select exactly TWO answers. [1 mark]

A. `CONCAT("A", "B")` produces "A B".
B. `SUBSTRING("STUDY", 2, 3)` produces "TUD".
C. $12 \text{ MOD } 4$ produces 0.
D. $9 / 2$ produces 4.

Answer: _____

6. What does `SUBSTRING("CAMPUS", 3, 2)` return under the supplied definition? [1 mark]

A. "MP"
B. "AM"
C. "MPU"
D. "CA"

Answer: _____

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Written responses | Questions 7-9

7. [4 marks]

Evaluate each expression. Show the operation performed first.

(a) $6 + 4 * 3$ (b) $(6 + 4) * 3$

(c) $15 / 4$ (d) $3 + 15 \text{ MOD } 4$

8. [4 marks]

There are 26 tiles and each full pack holds 7. State (a) the number of full packs, (b) $26 \text{ MOD } 7$, (c) the equality connecting these values, and (d) every possible remainder for a $\text{MOD } 7$ when a is a nonnegative integer.

9. [4 marks]

Use the supplied string procedures.

$s \leftarrow \text{"SUNRISE"}; \text{part} \leftarrow \text{SUBSTRING}(s, 4, 3); \text{tag} \leftarrow \text{CONCAT}(\text{part}, "!")$

Give (a) part , (b) tag , (c) the positions used, and (d) the final value of s .

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Written responses | Questions 10-12

10. [4 marks]

Trace: `minutes ← 75; hours ← minutes / 60; minutes ← 120.`

State (a) final hours, (b) final minutes, (c) why hours has that value, and (d) one assignment that updates hours to match the new minutes.

11. [4 marks]

The Workshop Pass displays `CONCAT("Hello, ", name)`, `SUBSTRING(name, 1, 2)`, `minutes / 60`, and `minutes MOD 60`. For `name = "Zoe"` and `minutes = 150`, give these four values in order.

12. [4 marks]

A learner uses `badge ← SUBSTRING(name, 0, 2)` and claims `/` always drops a fractional part. (a) Correct the assignment. (b) State the indexing rule. (c) Choose an integer minute value from 1-59 to test `minutes / 60`. (d) Give its expected result and explain what it checks.
